a **4-Hydroxyphthalic acid** — Disinfection by-products

Initial evidence names the compound among products formed during chlorination. [a1]

Criterion	Initial	Critique and evidence	After review
Formed from organic precursors	○ →	**Link formation to an organic precursor.** “The nonhalogenated aromatic DBPs contributed up to 84% of the formed monochloro-substitutes and 22% of the formed dichloro-substitutes, demonstrating that they somewhat acted as **intermediates between NOM and halogenated aromatic DBPs.”** [a1]	→ ✓

[a1] 10.1021/acs.est.9b06403

b **Isoterrelumamide A** — Microbial natural products

Initial evidence establishes fungal isolation and molecular structure. [b1]

Criterion	Initial	Critique and evidence	After review
Secondary-metabolite classification	○ →	**Establish secondary-metabolite status.** “In this study, Özkaya et al. isolated three new natural products Isopropylchaetominine (66), **Isoterrelumamide A (67),** and 5′-epi-Averufanin (68) (Figure 8), after inoculating the fungus in three different media and demonstrated how the production is depending on the composition of the culture medium in which the microorganism was inoculated; the **secondary metabolites** detected are not always the same [212].” [b2]	→ ✓

[b1] 10.1016/j.fitote.2018.10.008 [b2] 10.3390/md23020069

c **PC(20:2(11Z,14Z)/20:1(11Z))** — Endogenous metabolites

Initial evidence names PC(20:3/20:1), a different lipid from the target. [c1]

Criterion	Initial	Critique and evidence	After review
Exact target species PC(20:2/20:1)	○ →	**Find evidence for the exact target lipid.** “The Kennedy pathway represents the primary route for **phosphatidylcholine (PC) biosynthesis,** accounting for approximately 70–80% of total PC production in cells [21].” [c2]	→ ○

[c1] 10.1016/j.heliyon.2025.e41927 [c2] 10.3390/medicina61112057

✓ Explicit support in cited evidence ○ Not established by cited evidence